Data Communications

Lecture #3

What we have discussed

- Principles of network applications
- Web and HTTP

Today…

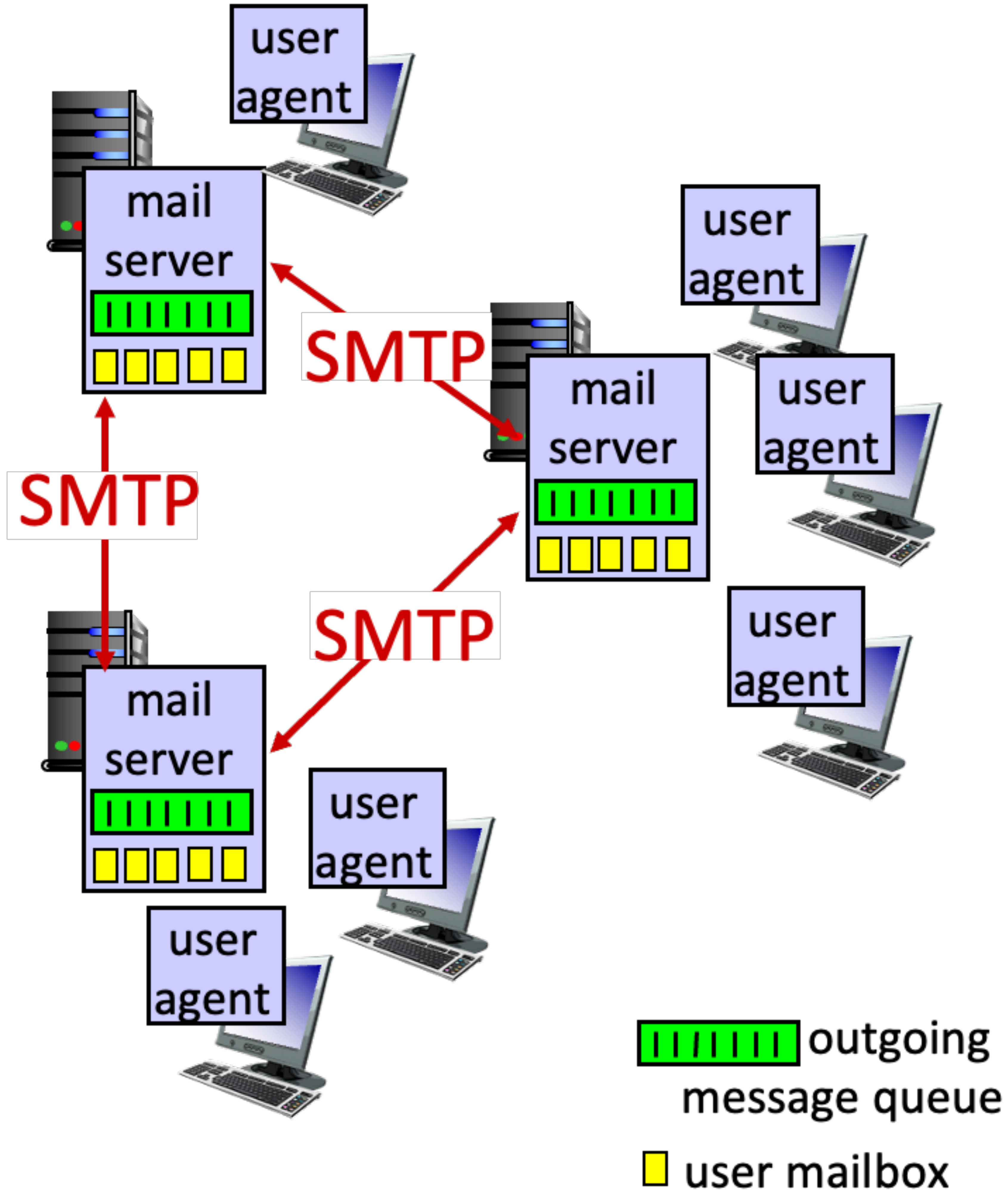
- E-mail, SMTP, IMAP
- Domain Name System (DNS)

E-mail, SMTP, IMAP

E-mail

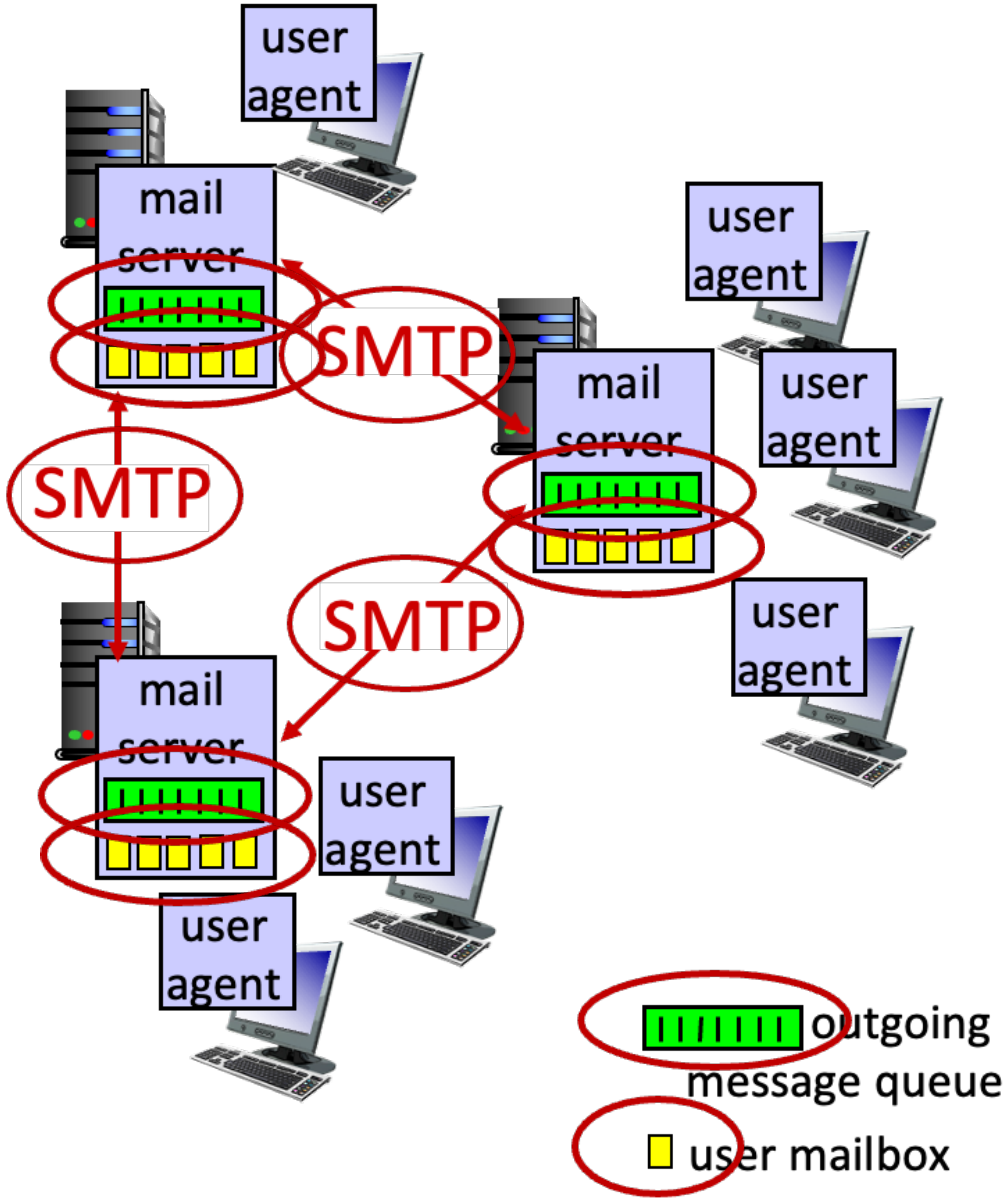
- Three major components
  - User agents
  - Mail servers
  - Simple Mail Transfer Protocol: SMTP

- User agent
  - a.k.a “mail reader”
  - Composing, editing, reading mail messages
    - e.g., Outlook, iPhone mail client
  - Outgoing, incoming messages stored on server



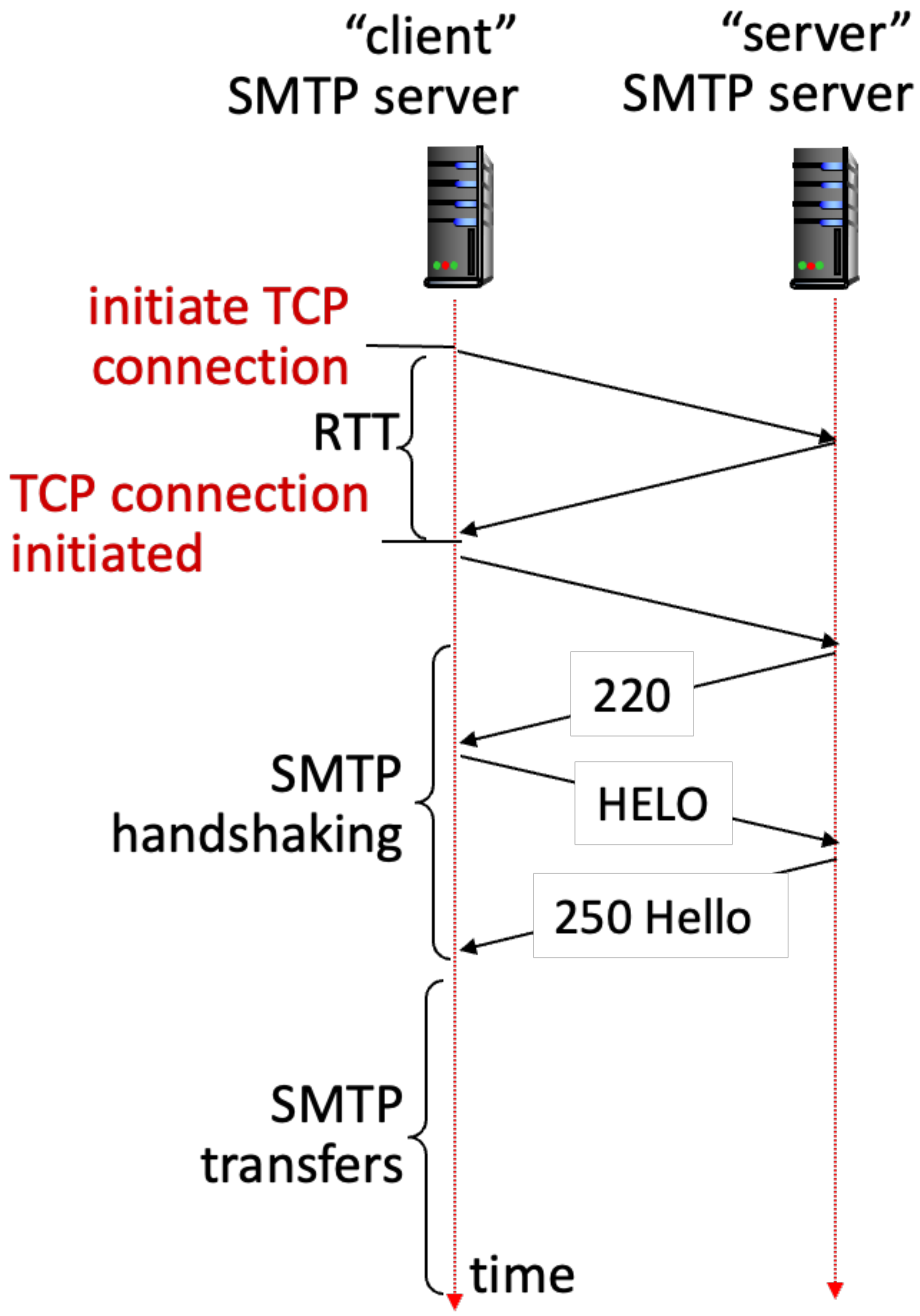
E-mail

- Mail servers
  - Mailbox contains incoming messages for user
  - Message queue of outgoing (to be sent) mail messages
- SMTP protocol between mail servers to send email messages
  - “client”: sending mail server
  - “server”: receiving mail server



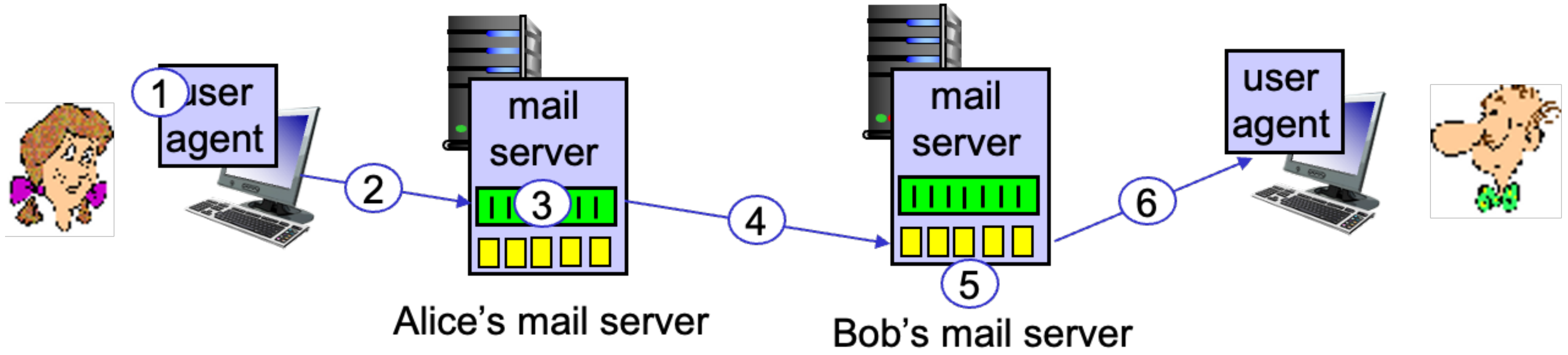
SMTP RFC (5321)

- Uses TCP to reliably transfer email message from client (mail server initiating connection) to server, port 25
  - Direct transfer: sending server (acting like client) to receiving server
- Three phases of transfer
  - SMTP handshaking (greeting)
  - SMTP transfer of messages
  - SMTP closure
- Command / response interaction (like HTTP)
  - Commands: ASCII text
  - Response: status code and phase



Example scenario

- Alice sends an email to Bob
  - Alice uses UA to compose e-mail message "to" bob@someschool.edu
  - Alice's UA sends message to her mail server using SMTP → message is placed in message queue
  - Client side of SMTP at mail server opens TCP connection with Bob's mail server
  - SMTP client sends Alice's message over the TCP connection
  - Bob's mail server places the message in Bob's mailbox
  - Bob invokes his user agent to read message



Sample SMTP interaction

```
% telnet hamburger.edu 25
  S: 220 hamburger.edu
```

Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
```

Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
```

Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
```

Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
```

Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
```

Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

SMTP observation: Comparison with HTTP

- HTTP: client pull
- SMTP: client push
- Both have ASCII command/response interaction, status code
- HTTP: each object is encapsulated in its own response message
- SMTP: multiple objects are sent in multipart message
- SMTP uses persistent connection
- SMTP requires message (header & body) to be in 7-bit ASCII
- SMTP server uses CRLF.CRLF to determine end of message

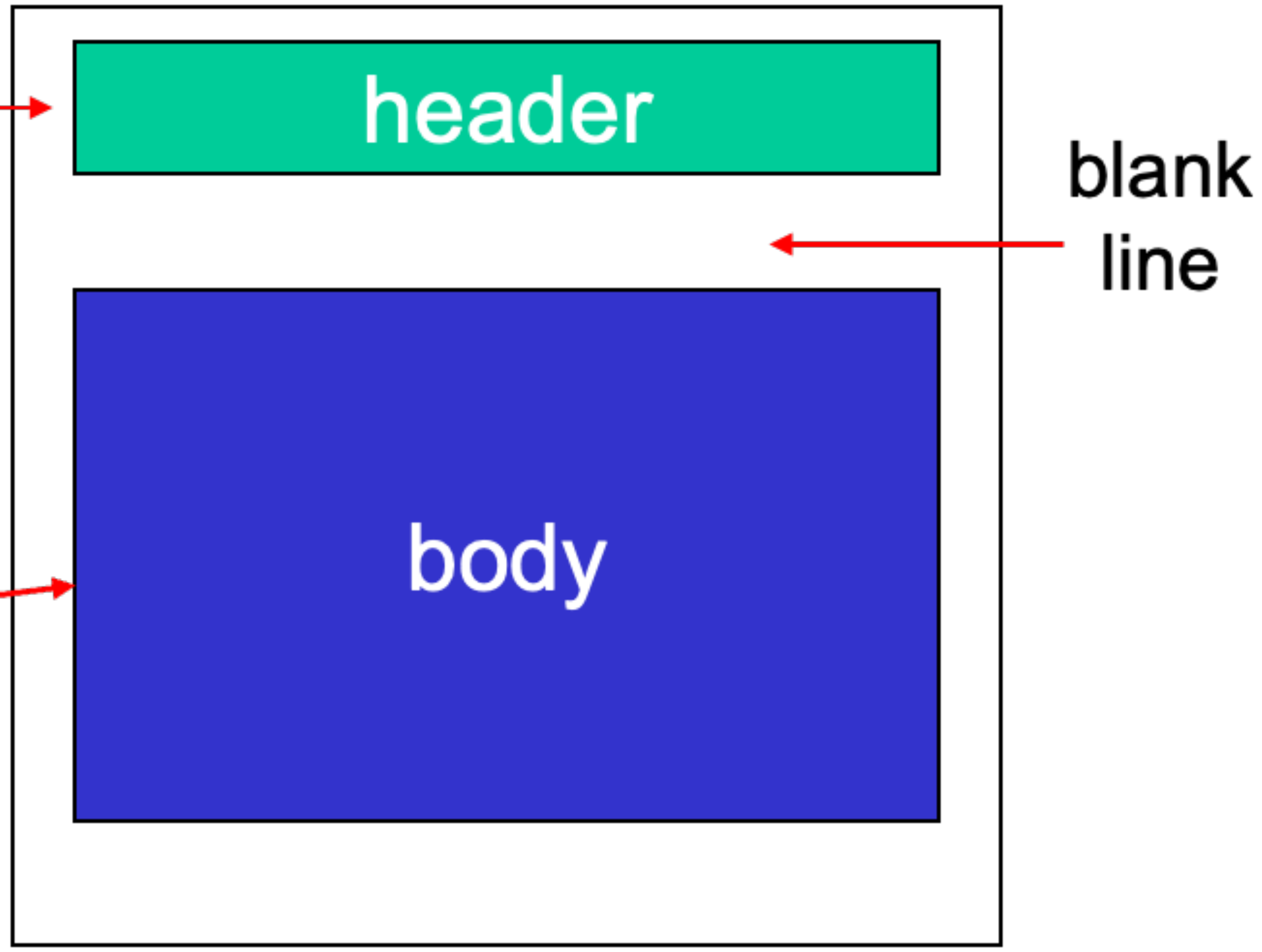
Mail message format

- SMTP: protocol for exchanging e-mail messages, defined in RFC 5321 (like RFC 7231 defines HTTP)
- RFC 2822 defines syntax for e-mail message itself (like HTML defines syntax for web documents)

Header line, e.g.,

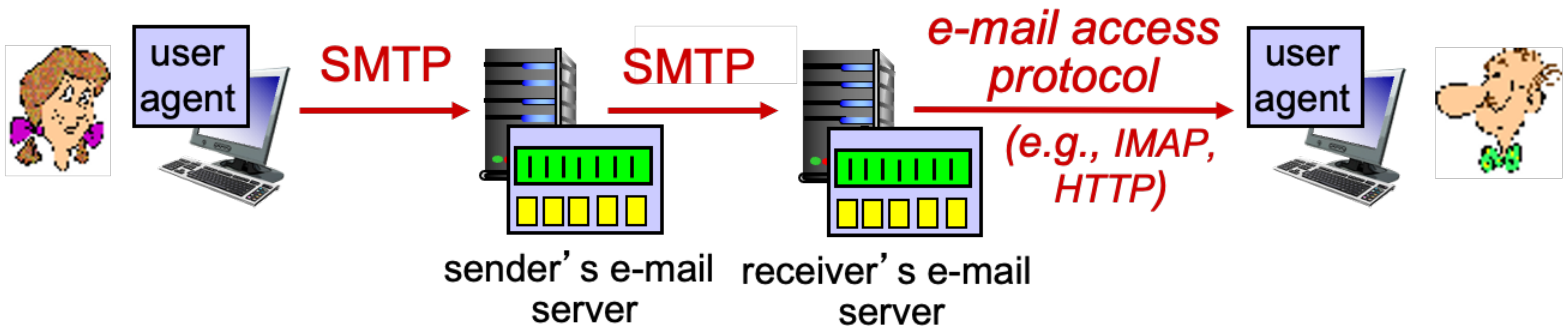
- To:
- From:
- Subject:

These lines, within the body of the email message area are different from SMTP MAIL FROM: RCPT TO: commands!



Body: the “message”, ASCII characters only

Retrieving email: mail access protocols



- SMTP: deliver/storage of e-mail messages to receiver’s server
- Mail access protocol: retrieval from server
    - IMAP: Internet Mail Access Protocol [RFC 3501]: messages stored on server, IMAP provides retrieval, deletion, folders of stored messages observer
- HTTP: gmail, hotmail, Yahoo!Mail, etc. provide web-based interface on top of SMTP (to send), IMAP (or POP) to retrieve e-mail messages

Domain Name System (DNS)

Domain Name System (DNS)

- People: many identifiers (e.g., SSN, name, passport#)

- Internet hosts, routers
  - IP address (32bit) - used for addressing datagrams
  - “Name”, e.g., cse.kangwon.ac.kr - used by humans

Q) How to map between IP address and name, and vice versa?

Domain Name System (DNS)

- DNS
  - Distributed database implemented in a hierarchy of many name servers
  - Application-layer protocol: hosts, DNS servers communicate to resolve names (address / name translation)
    - Note: core Internet function, implemented as application-layer protocol
    - Complexity at network's "edge"

DNS: services, structure

- DNS services
  - Hostname-to-IP address translation
  - Host aliasing
    - Canonical, alias names
  - Mail server aliasing
  - Load distribution
    - Replicated Web servers: many IP addresses correspond to one name

Q) Why not centralized DNS?

✓Single point of failure

✓Traffic volume

✓Distant centralized database

✓Maintenance

A) doesn’t scale!

✓Comcast DNS servers alone: 600B DNS queries/ day

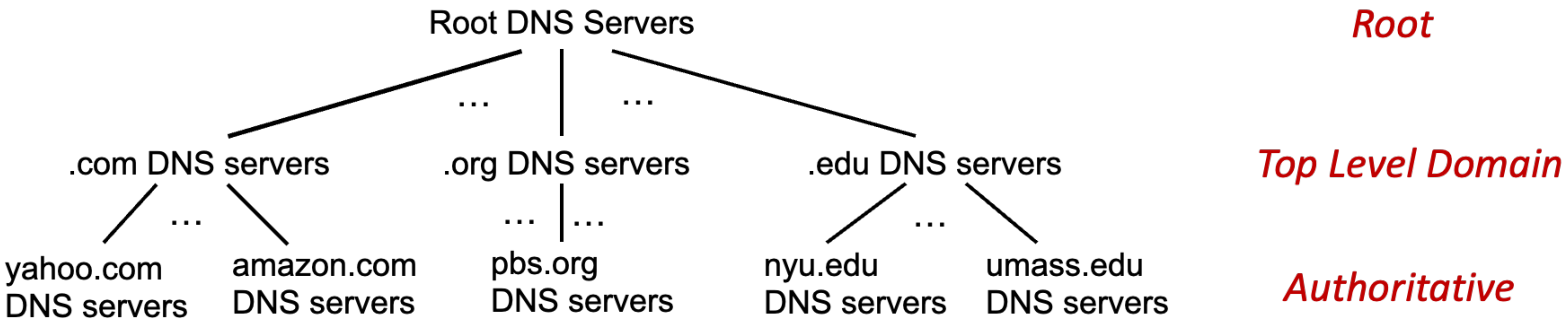
✓Akamai DNS servers alone: 2.2T DNS queries/day

Thinking about the DNS

- Humongous distributed database
  - ~billion records, each simple
- Handles many trillions of queries / day;
  - Many more reads than writes
  - Performance matters: almost every Internet transaction interacts with DNS - msecs count!
- Organizationally, physically decentralized
  - Millions of different organizations responsible for their records

→ "bulletproof": reliability, security

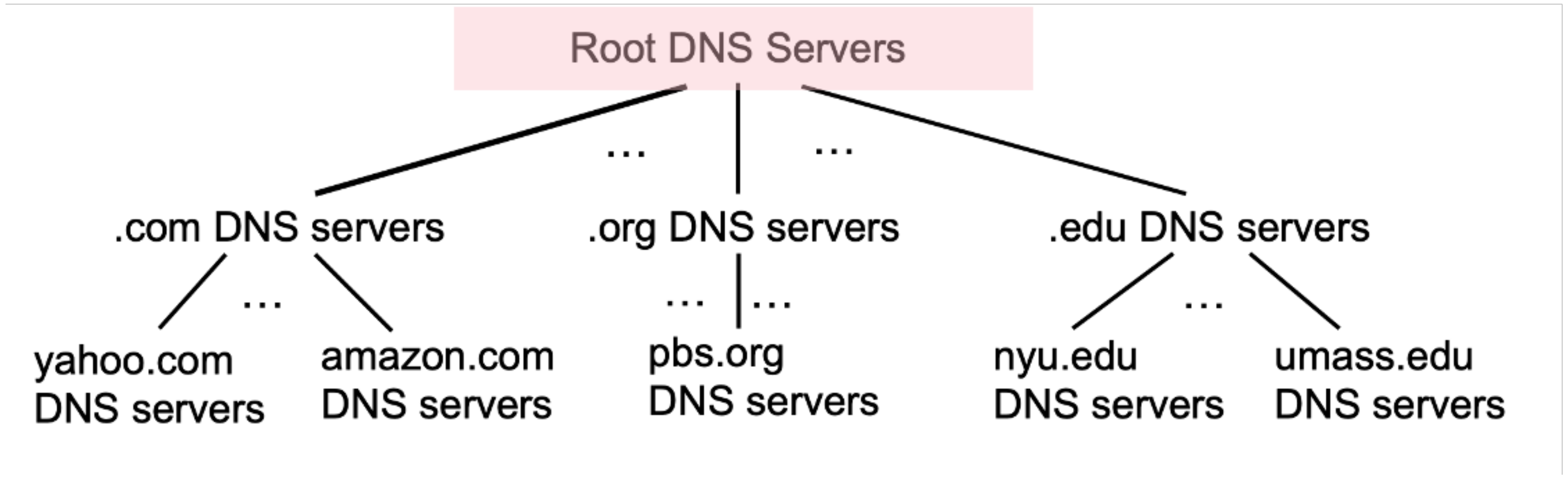
DNS: a distributed, hierarchical database



- Client wants IP address for www.amazon.com; 1^{st} approximation:
  - Client queries root server to find .com DNS server
  - Client queries .com DNS server to get amazon.com DNS server
  - Client queries amazon.com DNS server to get IP address for www.amazon.com

DNS: root name servers

- Official, contact-of-last-resort by name servers that can not resolve name
- Incredibly important Internet function
  - Internet couldn’t function without it
  - DNSSEC provides security for DNS services : authentication, message integrity

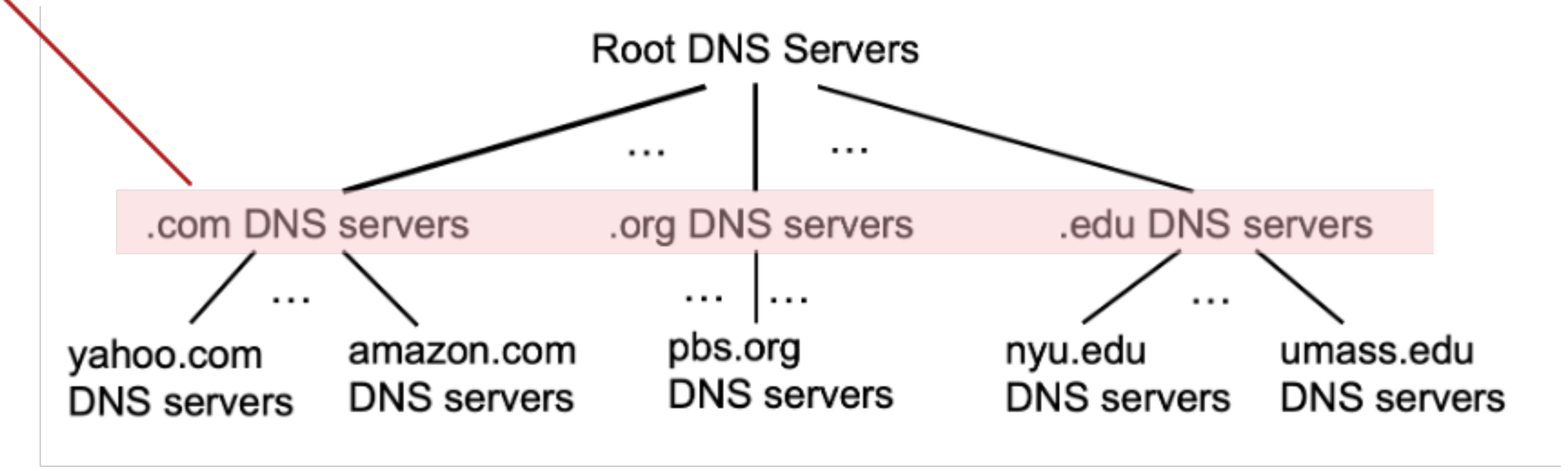


DNS: root name servers

- Official, contact-of-last-resort by name servers that can not resolve name
- Incredibly important Internet function
  - Internet couldn't function without it
  - DNSSEC provides security for DNS services : authentication, message integrity
- ICANN (Internet Corporation for Assigned Names and Numbers) manages root DNS domain
  - 13 logical root name "servers" worldwide
  - Each server is replicated many times
    - About 400 servers

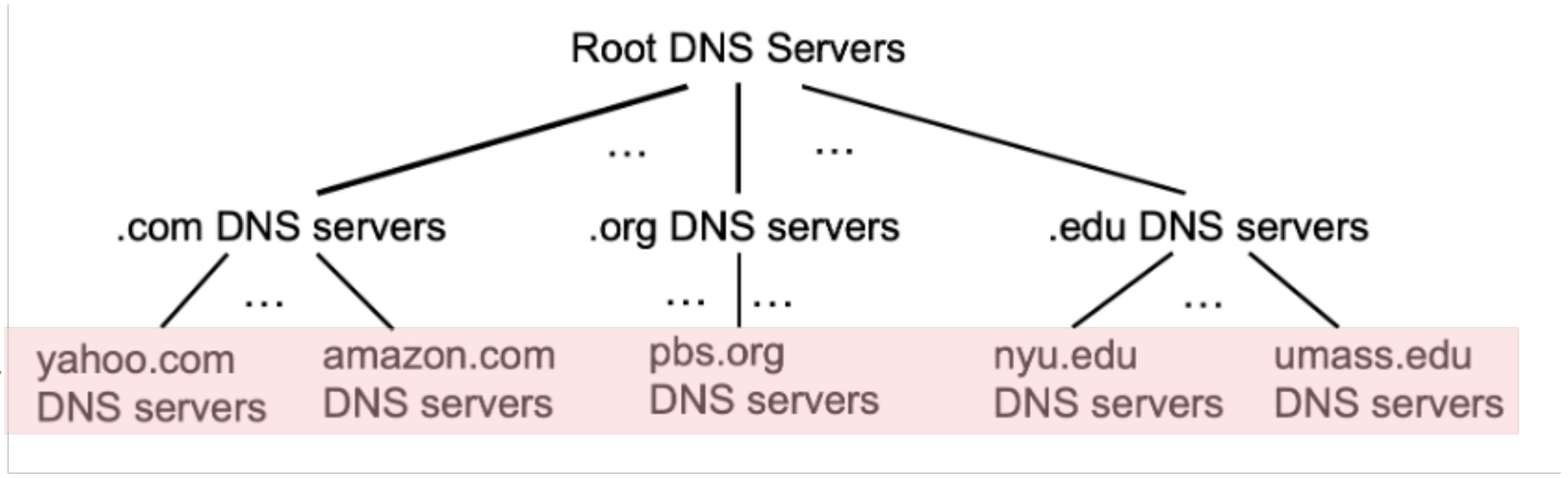
Top-Level Domain, and authoritative servers

- Top-Level Domain (TLD) servers
  - Responsible for .com, .org, .net, .edu, .aero, .jobs, .museums, and all top-level country domains, e.g., .kr, .cn, .uk, .fr, .ca, .jp
  - “Network Solutions”: authoritative registry for .com, .net TLD
  - “Educause”: .edu TLD



Top-Level Domain, and authoritative servers

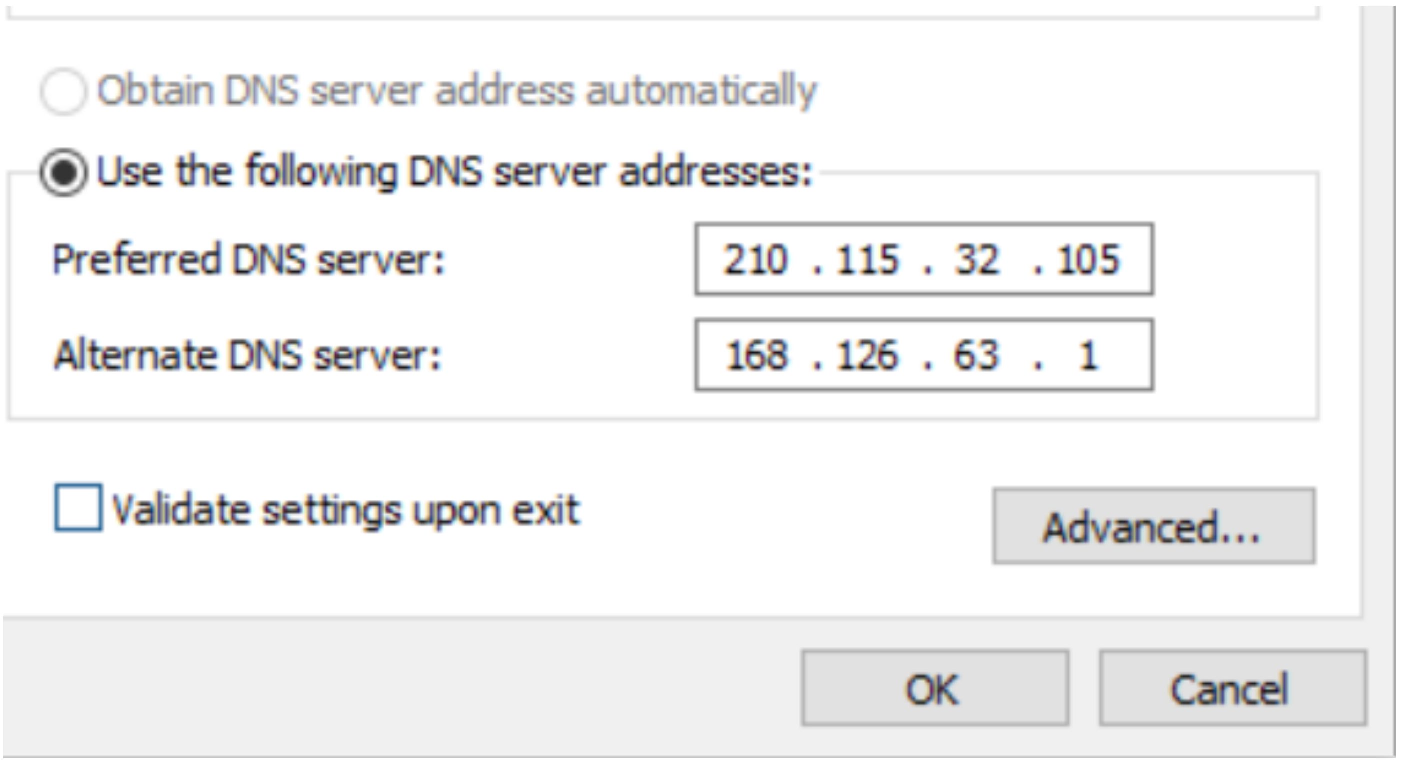
- Top-Level Domain (TLD) servers
  - Responsible for .com, .org, .net, .edu, .aero, .jobs, .museums, and all top-level country domains, e.g., .kr, .cn, .uk, .fr, .ca, .jp
  - “Network Solutions”: authoritative registry for .com, .net TLD
  - “Educause”: .edu TLD



- Authoritative DNS servers
  - Organization’s own DNS server(s), providing authoritative hostname to IP mappings for organization’s named hosts
  - Can be maintained by organization or service provider

Local DNS name servers

- When a host makes a DNS query, it is sent to its local DNS server
    - Local DNS server returns a reply, answering:
        - From its local cache of recent name-to-address translation pairs (possibly out of date!)
        - Forwarding the request into DNS hierarchy for resolution
    - Each ISP has its local DNS name server; to find yours
        - (MacOS) % scutil --dns
        - (Windows) > ipconfig /all



- Local DNS server doesn’t strictly belong to hierarchy

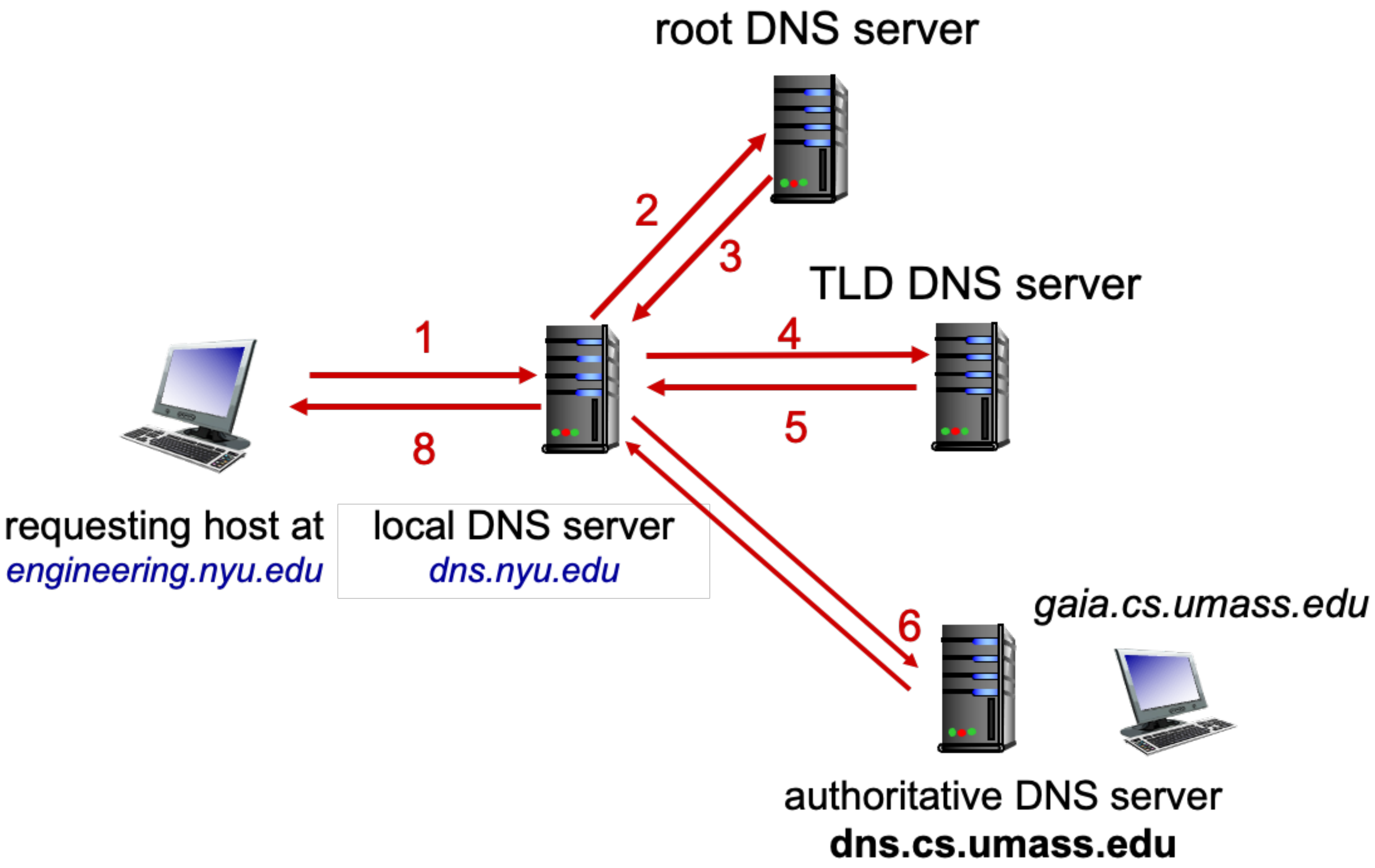
DNS name resolution: iterated query

- Example : host at engineering.nyu.edu wants IP address for gaia.cs.umass.edu

Iterative query

✓ Contacted server replies with name of server to contact

✓ "I don't know this name, but ask this server"



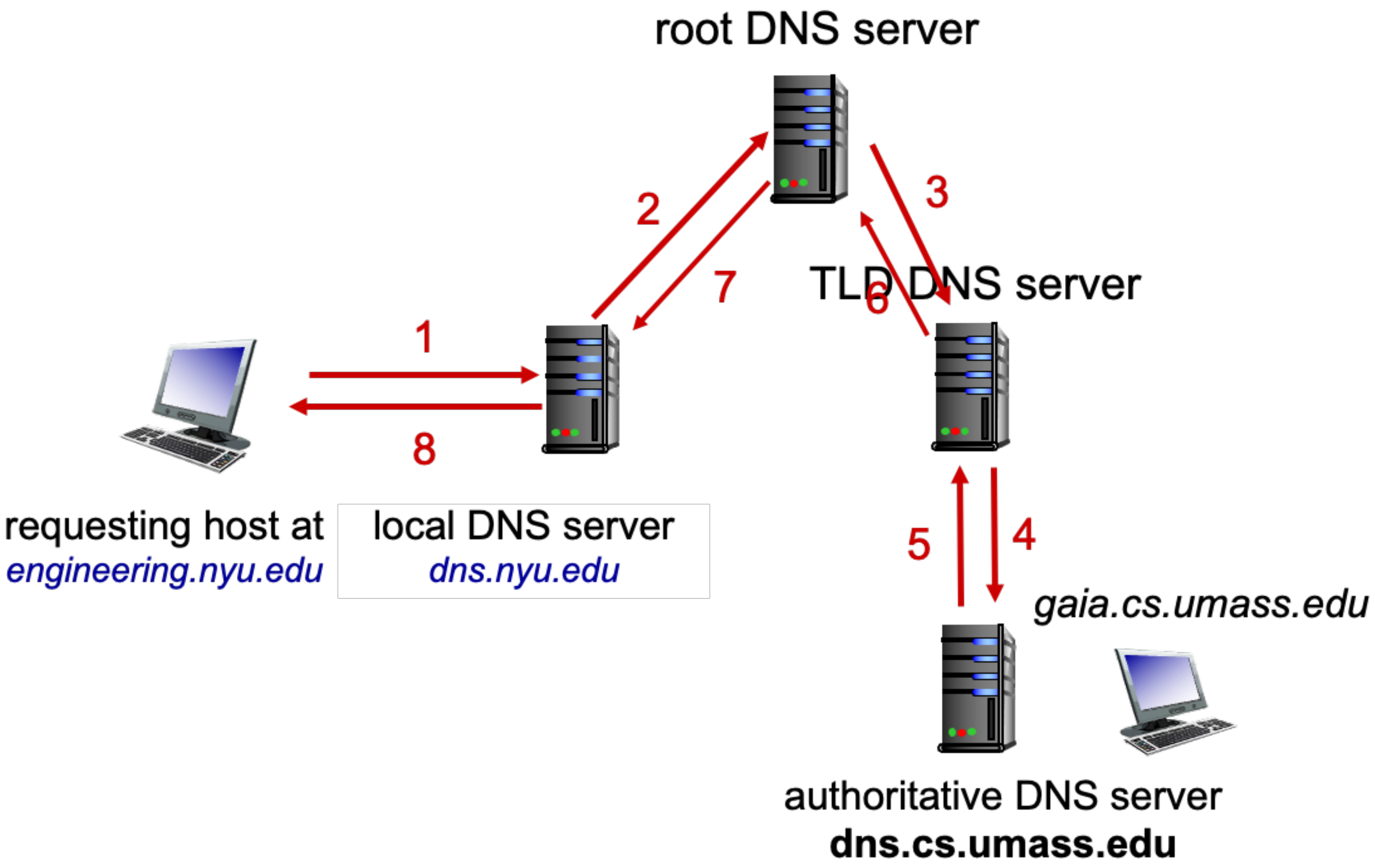
DNS name resolution: iterated query

- Example : host at engineering.nyu.edu wants IP address for gaia.cs.umass.edu

Recursive query

✓ Puts burden of name resolution on the contacted name server

✓ Heavy load at upper levels of hierarchy?



Caching DNS information

- Once (any) name server learns mapping, it caches mapping, and immediately returns a cached mapping in response to a query
  - Caching improves response time
  - Cache entries timeout (disappear) after some time (TTL)
  - TLD servers typically cached in local name servers
- Cached entries may be out-of-date
  - If named host changes IP address, may not be known Internet-wide until all TTLs expire!
  - Best-effort name-to-address translation!
  - Update/notify mechanisms proposed IETF standard (RFC2136)

DNS records

DNS: distributed database storing resource records (RR)

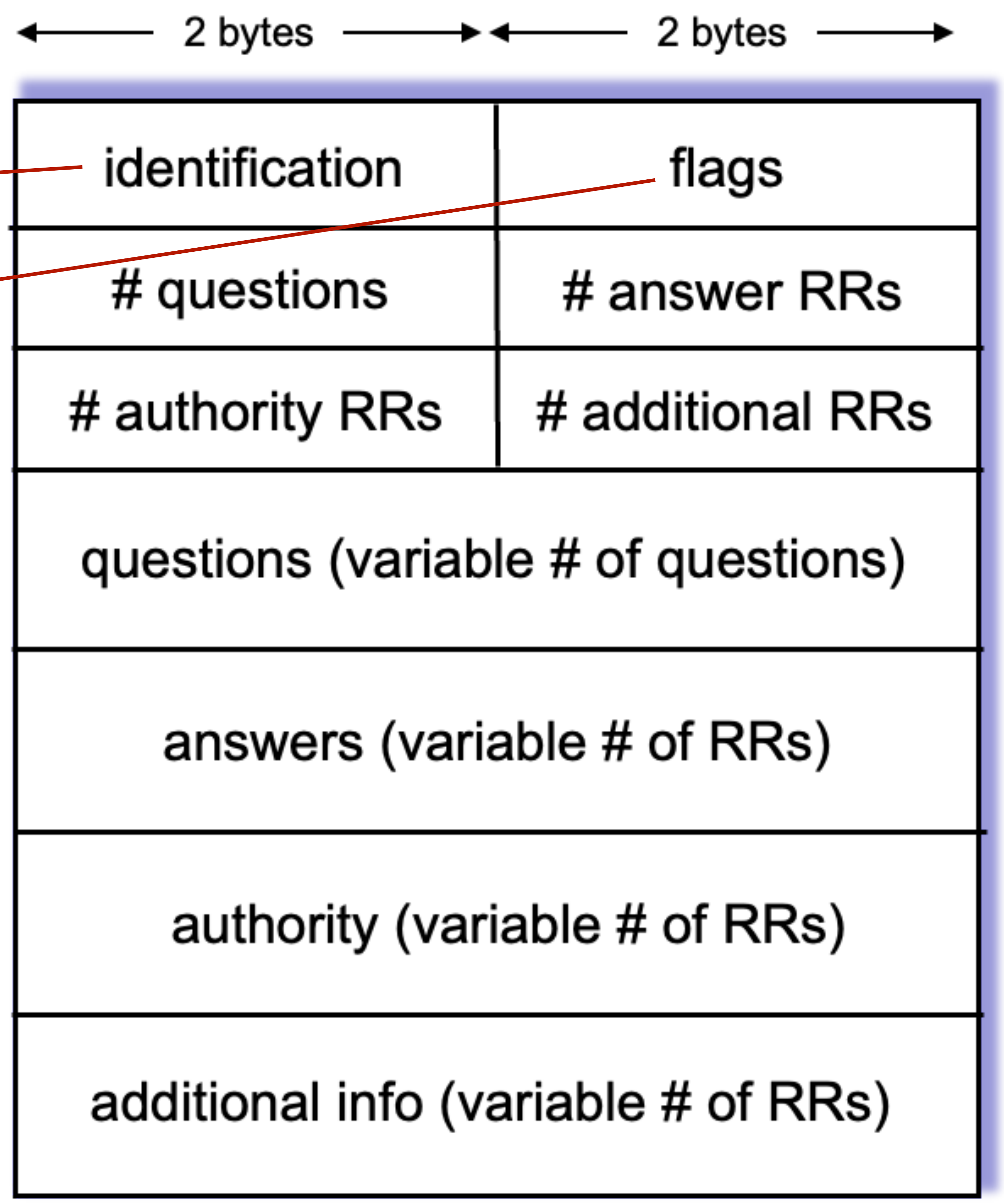
RR format: `(name, value, type, ttl)`

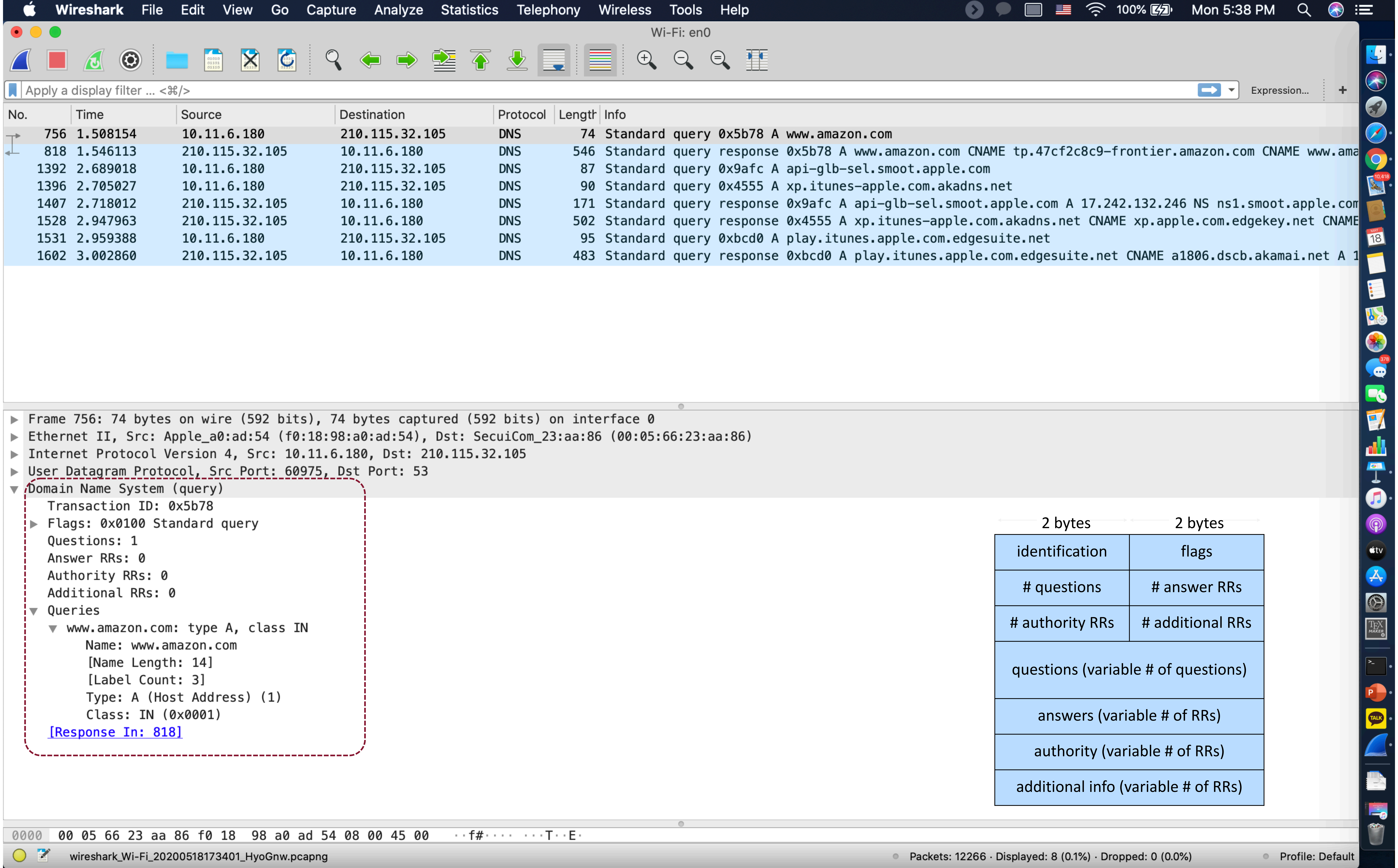
- type = A
  - name is **hostname**
  - value is **IP address**
- type = NS
  - name is **domain** (e.g., foo.com)
  - value is **host name of authoritative name server** for this domain
- type = CNAME
  - name is **alias name** for some “canonical” (the real) name
  - value is **canonical name**
- type = MX
  - name is **domain** (e.g., foo.com)
  - value is **host name of SMTP mail server** associated with the name

DNS protocol messages

- DNS query and reply messages, both have the same format

- Message header
  - identification: 16bit # for query, reply to query uses the same #
  - flags:
    - ✓ query or reply
    - ✓ Recursion desired
    - ✓ Recursion available
    - ✓ Reply is authoritative





Wireshark File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Wi-Fi: en0
Apply a display filter ... <⌘/>
Expression...
No. Time Source Destination Protocol Length Info
756 1.508154 10.11.6.180 210.115.32.105 DNS 74 Standard query 0x5b78 A www.amazon.com
818 1.546113 210.115.32.105 10.11.6.180 DNS 546 Standard query response 0x5b78 A www.amazon.com CNAME tp.47cf2c8c9-frontier.amazon.com CNAME www.ama
1392 2.689018 10.11.6.180 210.115.32.105 DNS 87 Standard query 0x9afc A api-glb-sel.smoot.apple.com
1396 2.705027 10.11.6.180 210.115.32.105 DNS 90 Standard query 0x4555 A xp.itunes-apple.com.akadns.net
1407 2.718012 210.115.32.105 10.11.6.180 DNS 171 Standard query response 0x9afc A api-glb-sel.smoot.apple.com A 17.242.132.246 NS ns1.smoot.apple.com
1528 2.947963 210.115.32.105 10.11.6.180 DNS 502 Standard query response 0x4555 A xp.itunes-apple.com.akadns.net CNAME xp.apple.com.edgekey.net CNAME
1531 2.959388 10.11.6.180 210.115.32.105 DNS 95 Standard query 0xbcd0 A play.itunes.apple.com.edgesuite.net
1602 3.002860 210.115.32.105 10.11.6.180 DNS 483 Standard query response 0xbcd0 A play.itunes.apple.com.edgesuite.net CNAME a1806.dscb.akamai.net A 1
Frame 756: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
Ethernet II, Src: Apple_a0:ad:54 (f0:18:98:a0:ad:54), Dst: SecuiCom_23:aa:86 (00:05:66:23:aa:86)
Internet Protocol Version 4, Src: 10.11.6.180, Dst: 210.115.32.105
User Datagram Protocol, Src Port: 60975, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x5b78
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
www.amazon.com: type A, class IN
Name: www.amazon.com
[Name Length: 14]
[Label Count: 3]
Type: A (Host Address) (1)
Class: IN (0x0001)
[Response In: 818]
2 bytes
2 bytes
identification
flags
questions
answer RRs
authority RRs
additional RRs
questions (variable # of questions)
answers (variable # of RRs)
authority (variable # of RRs)
additional info (variable # of RRs)
0000 00 05 66 23 aa 86 f0 18 98 a0 ad 54 08 00 45 00 ··f#···· ···T··E·
wireshark_Wi-Fi_20200518173401_HyoGnw.pcapng
Packets: 12266 · Displayed: 8 (0.1%) · Dropped: 0 (0.0%)
Profile: Default

DNS protocol messages

- DNS query and reply messages, both have the same format

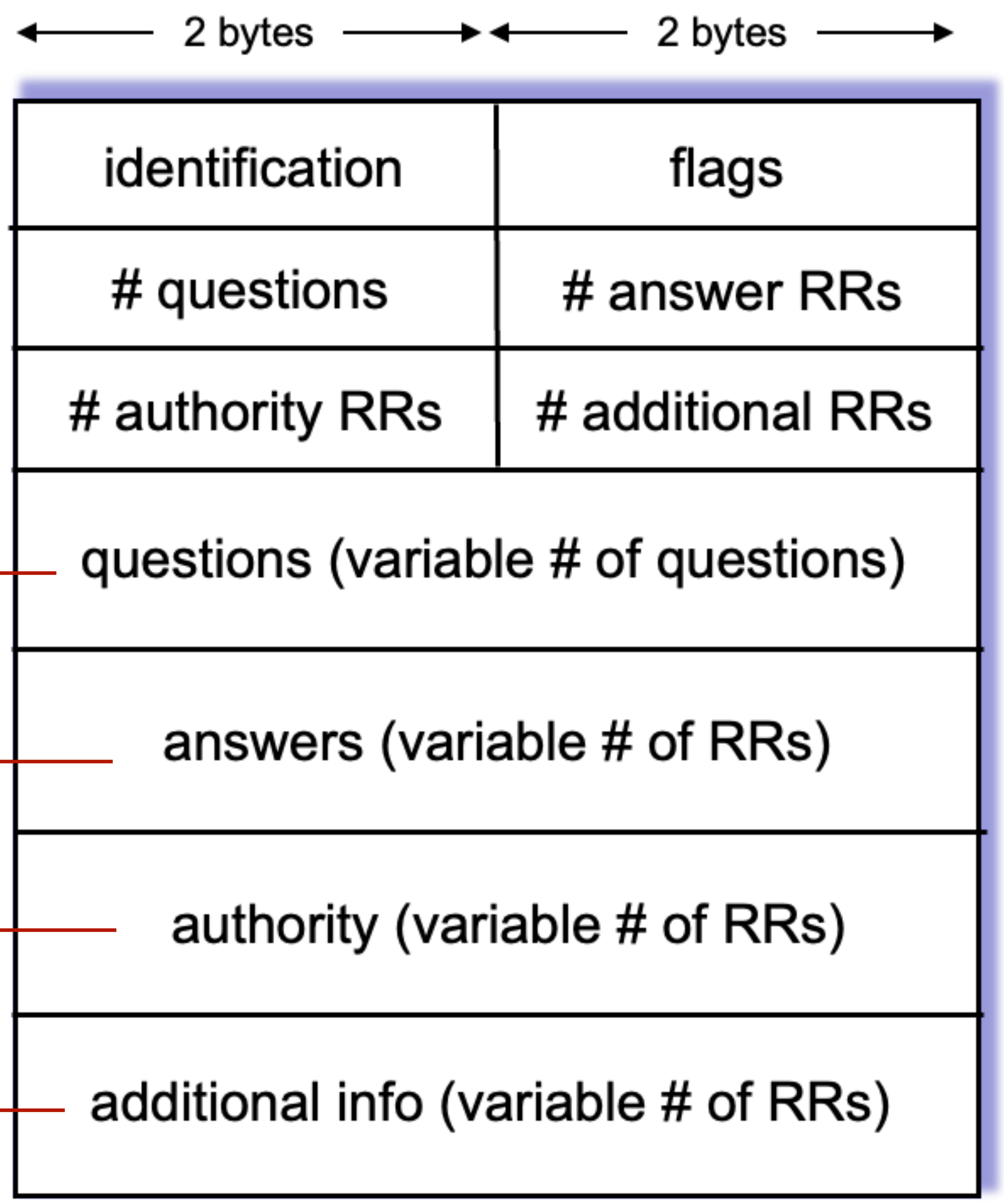
- Message header

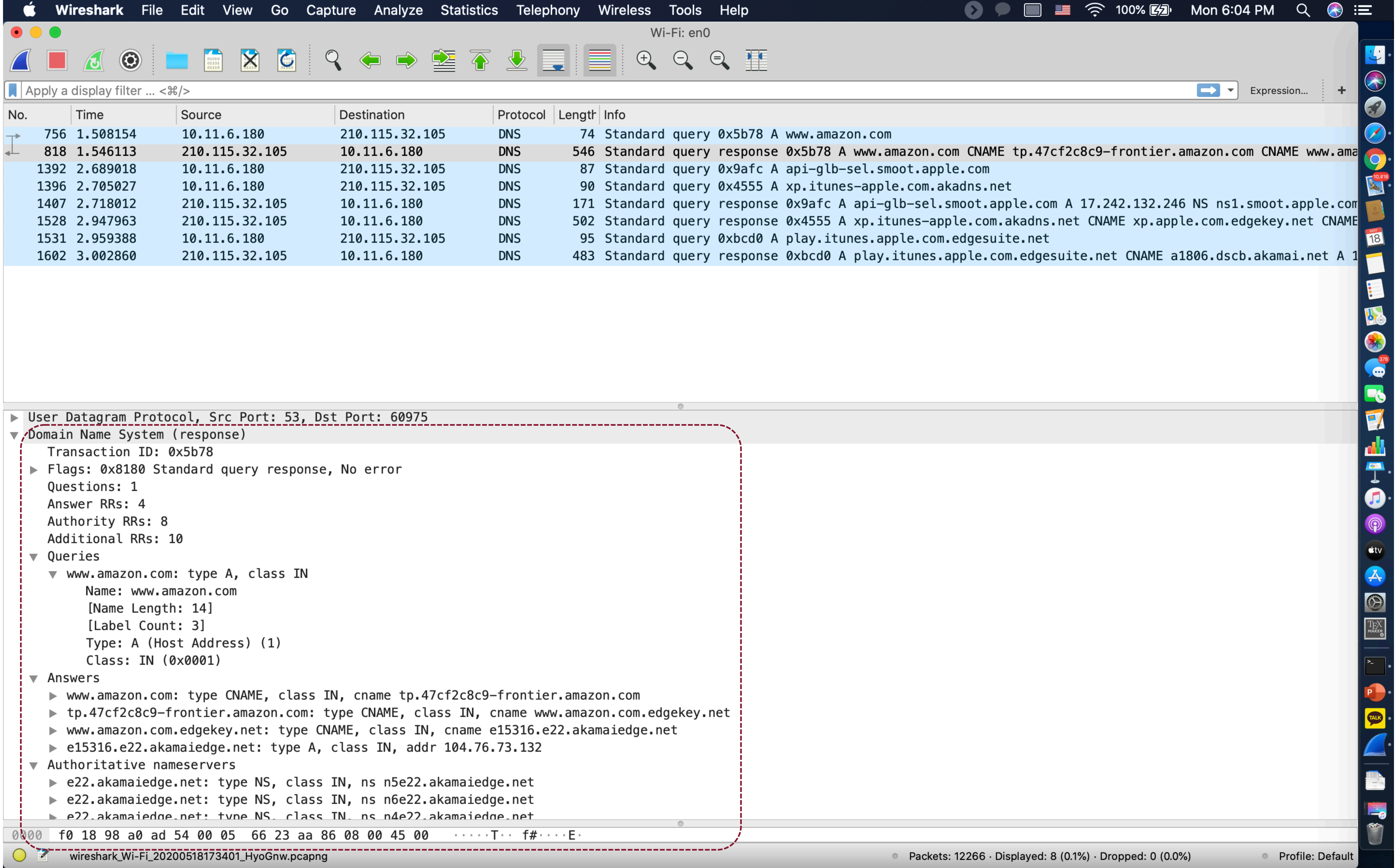
  - name, type fields for a query

  - RRs in response to query

  - records for authoritative servers

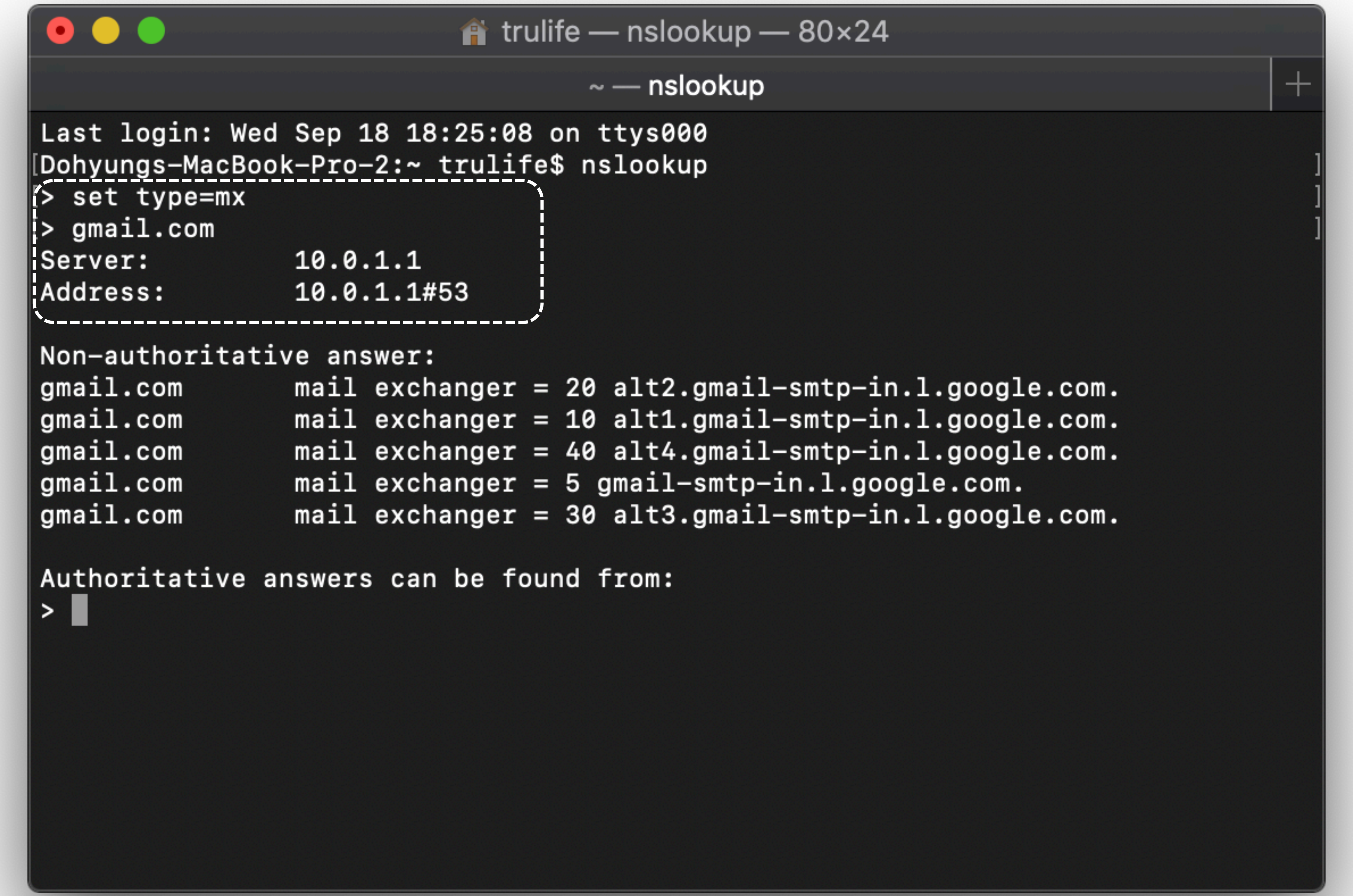
  - additional “helpful” info that may be used




Wireshark File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
100% Mon 6:04 PM
Wi-Fi: en0
Apply a display filter ... <⌘/>
Expression...
No. Time Source Destination Protocol Length Info
756 1.508154 10.11.6.180 210.115.32.105 DNS 74 Standard query 0x5b78 A www.amazon.com
818 1.546113 210.115.32.105 10.11.6.180 DNS 546 Standard query response 0x5b78 A www.amazon.com CNAME tp.47cf2c8c9-frontier.amazon.com CNAME www.ama
1392 2.689018 10.11.6.180 210.115.32.105 DNS 87 Standard query 0x9afc A api-glb-sel.smoot.apple.com
1396 2.705027 10.11.6.180 210.115.32.105 DNS 90 Standard query 0x4555 A xp.itunes-apple.com.akadns.net
1407 2.718012 210.115.32.105 10.11.6.180 DNS 171 Standard query response 0x9afc A api-glb-sel.smoot.apple.com A 17.242.132.246 NS ns1.smoot.apple.com
1528 2.947963 210.115.32.105 10.11.6.180 DNS 502 Standard query response 0x4555 A xp.itunes-apple.com.akadns.net CNAME xp.apple.com.edgekey.net CNAME
1531 2.959388 10.11.6.180 210.115.32.105 DNS 95 Standard query 0xbcd0 A play.itunes.apple.com.edgesuite.net
1602 3.002860 210.115.32.105 10.11.6.180 DNS 483 Standard query response 0xbcd0 A play.itunes.apple.com.edgesuite.net CNAME a1806.dscb.akamai.net A 1
User Datagram Protocol, Src Port: 53, Dst Port: 60975
Domain Name System (response)
Transaction ID: 0x5b78
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 4
Authority RRs: 8
Additional RRs: 10
Queries
www.amazon.com: type A, class IN
Name: www.amazon.com
[Name Length: 14]
[Label Count: 3]
Type: A (Host Address) (1)
Class: IN (0x0001)
Answers
www.amazon.com: type CNAME, class IN, cname tp.47cf2c8c9-frontier.amazon.com
tp.47cf2c8c9-frontier.amazon.com: type CNAME, class IN, cname www.amazon.com.edgekey.net
www.amazon.com.edgekey.net: type CNAME, class IN, cname e15316.e22.akamaiedge.net
e15316.e22.akamaiedge.net: type A, class IN, addr 104.76.73.132
Authoritative nameservers
e22.akamaiedge.net: type NS, class IN, ns n5e22.akamaiedge.net
e22.akamaiedge.net: type NS, class IN, ns n6e22.akamaiedge.net
e22.akamaiedge.net: type NS, class IN, ns n4e22.akamaiedge.net
0000 f0 18 98 a0 ad 54 00 05 66 23 aa 86 08 00 45 00 ·····T·· f#····E·
wireshark_Wi-Fi_20200518173401_HyoGnw.pcapng
Packets: 12266 · Displayed: 8 (0.1%) · Dropped: 0 (0.0%)
Profile: Default

Example of DNS process

- How to send a mail to mr.dhkim@gmail.com
  - Lookup mx record to gmail.com
  - Lookup a record of a host in the response of the above



From IP address to name

- RR type : PTR
- **in-addr.arpa** domain
- e.g., $nslookup 210.115.32.105

```
albert@Dohyungs-iMac ~ % nslookup 210.115.32.105
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
105.32.115.210.in-addr.arpa     name = ns.kangwon.ac.kr.

Authoritative answers can be found from:

albert@Dohyungs-iMac ~ %
```

Getting your info into the DNS

- Suppose you launch a new startup “Network Utopia”
  - Register name "networkuptopia.com" at DNS registrar (e.g., Network Solutions)
    - You need to provide name/IP address of authoritative name servers (primary and secondary)
    - DNS registrar verifies the uniqueness of the name
    - DNS registrar inserts two RRs (NS, A) into .com TLD server
      (networkutopia.com, dns1.networkutopia.com, **NS**)
      (dns1.networkutopia.com, 212.212.212.1, **A**)
  - Create authoritative server locally with IP address 212.212.212.1
    - Type A record for www.networkutopia.com
    - Type MX record for networkutopia.com

DNS security

- DDoS attacks
  - Bombard root servers with traffic
    - Not successful to date; traffic filtering
    - Local DNS servers cache IPs of TLD servers, allowing root server bypass
  - Bombard TLD servers
    - Potentially more dangerous
- Spoofing attacks
  - Intercept DNS queries, returning bogus replies
    - DNS cache poisoning
    - RFC 4033: DNSSEC authentication services

Refer to the following link for DNS relevant attacks: https://kameinao.medium.com/dangers-threatening-tld-dns-servers-a-looming-digital-crisis-3b3cf459e82e

Questions?